

6.2 (a) From the property of multi-variable Dirac function,

$$\delta(\vec{f}(\vec{x})) = \frac{\delta(\vec{x} - \vec{x}_0)}{|\det[J\vec{f}(\vec{x}_0)]|},$$

where $\vec{x}_0$ is the solution to the vector equation $\vec{f}(\vec{x}_0) = 0$ and $J\vec{f}(\vec{x}_0)$ is the Jacobian of $\vec{f}$ at $\vec{x} = \vec{x}_0$.

In this case, $\vec{f}(\vec{x}) = \vec{x}' - \vec{r}(t - \frac{|\vec{x} - \vec{x}'|}{c})$, and the Jacobian of this function is

$$J\vec{f}(\vec{x}') = \begin{pmatrix} 1 - v_x n_x/c & -v_x n_y/c & -v_x n_z/c \\ -v_y n_x/c & 1 - v_y n_y/c & -v_y n_z/c \\ -v_z n_x/c & -v_z n_y/c & 1 - v_z n_z/c \end{pmatrix}$$

Here $\vec{n} = (\vec{x} - \vec{x}')/|\vec{x} - \vec{x}'|$ and $\vec{v}$ is the velocity of the particle evaluated at the retarded time.

By direct calculation, the determinant of the Jacobian is

$$\det[J\vec{f}(\vec{x}')] = 1 - \vec{v} \cdot \vec{n}/c.$$

Denote the solution of $\vec{f}(\vec{x}') = 0$ as $\vec{x}'_0$, then

$$\int \delta[\vec{x} - \vec{r}(t - \frac{|\vec{x} - \vec{x}'|}{c})] d^3x' = \int \frac{\delta(\vec{x}' - \vec{x}'_0)}{|\det[J\vec{f}(\vec{x}'_0)]|} d^3x' = \frac{1}{1 - \vec{v} \cdot \vec{n}/c} = \frac{1}{k},$$

where $k = 1 - \vec{v} \cdot \vec{n}/c$, $\vec{v}$ and $\vec{n}$ being evaluated at $\vec{x}' = \vec{x}'_0$ and the retarded time.

(b) Using Eq. (6.55), the electric field is ($\hat{R}$ is the same as $\vec{n}$ in part (a) and $R = |\vec{x} - \vec{x}'|$)

$$\vec{E}(\vec{x}, t) = \frac{1}{4\pi\epsilon_0} \int d^3x' \left\{ \frac{\hat{R}}{R^2} [\rho(\vec{x}', t')]_{\text{ret}} + \frac{\hat{R}}{cR} \left[\frac{\partial \rho(\vec{x}', t')}{\partial t'} \right]_{\text{ret}} - \frac{1}{c^2 R} \left[\frac{\partial \vec{J}(\vec{x}', t')}{\partial t'} \right]_{\text{ret}} \right\}$$

As pointed out by Eq. (6.57), $\left[\frac{\partial f(\vec{x}', t')}{\partial t'} \right]_{\text{ret}} = \frac{\partial}{\partial t} [f(\vec{x}', t')]_{\text{ret}}$, the above expression can be written as, also noticing that R does not depend on t ,

$$\vec{E}(\vec{x}, t) = \frac{1}{4\pi\epsilon_0} \int d^3x' \left\{ \left[\frac{\hat{R}}{R^2} \rho(\vec{x}', t') \right]_{\text{ret}} + \frac{1}{c} \frac{\partial}{\partial t} \left[\frac{\hat{R}}{R} \rho(\vec{x}', t') \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[\frac{\vec{J}(\vec{x}', t')}{R} \right]_{\text{ret}} \right\}$$

Since $\rho(\vec{x}', t') = q \delta(\vec{x}' - \vec{r}(t_{\text{ret}}))$ and $\vec{J} = \rho \vec{v}(t')$, the integration will lead to an extra retardation factor as in part (a). Therefore,

$$\vec{E}(\vec{x}, t) = \frac{q}{4\pi\epsilon_0} \left\{ \left[\frac{\hat{R}}{k R^2} \right]_{\text{ret}} + \frac{1}{c} \frac{\partial}{\partial t} \left[\frac{\hat{R}}{k R} \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[\frac{\vec{v}}{k R} \right]_{\text{ret}} \right\}.$$

Following the same procedure, using Eq. (6.56),

$$\begin{aligned}\vec{B}(\vec{x}, t) &= \frac{\mu_0}{4\pi} \int d^3x' \left\{ \left[\rho \vec{v} \times \frac{\hat{R}}{R} \right]_{\text{ret}} + \frac{1}{c} \frac{\partial}{\partial t} \left[\rho \vec{v} \times \frac{\hat{R}}{R} \right]_{\text{ret}} \right\} \\ &= \frac{\mu_0 q}{4\pi} \left\{ \left[\frac{\vec{v} \times \hat{R}}{R^2} \right]_{\text{ret}} + \frac{1}{c} \frac{\partial}{\partial t} \left[\frac{\vec{v} \times \hat{R}}{R} \right]_{\text{ret}} \right\}\end{aligned}$$

(c) Let us first establish a few useful relations. Denote $\vec{R} = \vec{x} - \vec{x}'(t')$, then

$$\frac{\partial}{\partial t'} R = \frac{\partial}{\partial t'} |\vec{x} - \vec{x}'(t')| = \frac{1}{|\vec{x} - \vec{x}'(t')|} (\vec{x} - \vec{x}'(t')) \cdot (-\vec{v}(t')) = -\vec{v}(t') \cdot \hat{R},$$

where $\vec{v} = \vec{x}'$ and $\hat{R} = (\vec{x} - \vec{x}'(t'))/|\vec{x} - \vec{x}'(t')|$. By applying chain rule, we will have

$$\frac{\partial}{\partial t'} \left(\frac{1}{R} \right) = -\frac{1}{R^2} \frac{\partial}{\partial t'} R = \frac{\vec{v}(t') \cdot \hat{R}}{R^2}, \quad \text{and}$$

$$\frac{\partial}{\partial t'} \hat{R} = \frac{\partial}{\partial t'} \left(\frac{\vec{R}}{R} \right) = -\frac{\vec{v}(t')}{R} + \frac{(\vec{v}(t') \cdot \hat{R}) \hat{R}}{R}. \quad \text{Also, for retarded time, } t' = t - |\vec{x} - \vec{x}'(t')|/c, \text{ from}$$

which we can obtain $\frac{dt}{dt'} = 1 - \frac{1}{c} \frac{d}{dt'} R = 1 - \frac{\vec{v}(t') \cdot \hat{R}}{c}$. Now, consider the Jefimenko expression for

the magnetic field, The second term can be written as

$$\frac{\partial}{\partial t} \left[\frac{\vec{v} \times \hat{R}}{R} \right]_{\text{ret}} = \frac{1}{[R]_{\text{ret}}} \frac{\partial}{\partial t} \left[\frac{\vec{v} \times \hat{R}}{R} \right] + \left[\frac{\vec{v} \times \hat{R}}{R} \right]_{\text{ret}} \frac{\partial}{\partial t} \left[\frac{1}{R} \right]_{\text{ret}}$$

$$\text{Since } \frac{\partial}{\partial t} \left[\frac{1}{R} \right]_{\text{ret}} = \frac{\partial}{\partial t'} \left[\frac{1}{R} \right]_{\text{ret}} \frac{\partial t'}{\partial t} = \frac{1}{R} \frac{\partial}{\partial t'} \left(\frac{1}{R(t')} \right) = \frac{\vec{v}(t') \cdot \hat{R}}{R^2 R(t')^2} = \left[\frac{\vec{v} \cdot \hat{R}}{R^3} \right]_{\text{ret}},$$

$$\begin{aligned}\text{then } \vec{B}(\vec{x}, t) &= \frac{\mu_0 q}{4\pi} \left\{ \left[\frac{\vec{v} \times \hat{R}}{R^2} \right]_{\text{ret}} + \frac{1}{c [R]_{\text{ret}}} \frac{\partial}{\partial t} \left[\frac{\vec{v} \times \hat{R}}{R} \right]_{\text{ret}} + \frac{1}{c} \left[\frac{\vec{v} \times \hat{R}}{R} \cdot \frac{\vec{v} \cdot \hat{R}}{R^2} \right]_{\text{ret}} \right\} \\ &= \frac{\mu_0 q}{4\pi} \left\{ \left[\frac{\vec{v} \times \hat{R}}{R^2} \right]_{\text{ret}} \left[1 + \frac{\vec{v} \cdot \hat{R}}{Rc} \right]_{\text{ret}} + \frac{1}{c [R]_{\text{ret}}} \frac{\partial}{\partial t} \left[\frac{\vec{v} \times \hat{R}}{R} \right]_{\text{ret}} \right\} \\ &= \frac{\mu_0 q}{4\pi} \left\{ \left[\frac{\vec{v} \times \hat{R}}{R^2} \right]_{\text{ret}} + \frac{1}{c [R]_{\text{ret}}} \frac{\partial}{\partial t} \left[\frac{\vec{v} \times \hat{R}}{R} \right]_{\text{ret}} \right\}.\end{aligned}$$

where we have used $1 + \vec{v} \cdot \hat{R}/Rc = 1/R$

To verify the equivalence for electric field, we start from the Feynman's expression. Notice that

$$\begin{aligned}
 \frac{1}{c^2} \frac{\partial^2}{\partial t^2} [\hat{R}]_{\text{ret}} &= \frac{1}{c^2} \frac{\partial}{\partial t} \left[\frac{1}{R} \frac{\partial}{\partial t} (\hat{R}) \right]_{\text{ret}} = \frac{1}{c^2} \frac{\partial}{\partial t} \left[\frac{(\vec{v} \cdot \hat{R}) \hat{R} - \vec{v}}{KR} \right]_{\text{ret}} \\
 &= \frac{1}{c^2} \frac{\partial}{\partial t} \left[\frac{(\vec{v} \cdot \hat{R}) \hat{R}}{KR} \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[\frac{\vec{v}}{KR} \right]_{\text{ret}} \\
 &= \frac{1}{c^2} \frac{\partial}{\partial t} \left[\frac{\hat{R}}{KR} \left(-(1 - \vec{v} \cdot \hat{R}/c) + 1 \right) \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[\frac{\vec{v}}{KR} \right]_{\text{ret}} \\
 &= \frac{1}{c^2} \frac{\partial}{\partial t} \left[\frac{\hat{R}}{KR} \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[\frac{\hat{R}}{R} \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[\frac{\vec{v}}{KR} \right]_{\text{ret}}
 \end{aligned}$$

$$\begin{aligned}
 \text{and } \frac{[R]_{\text{ret}}}{c} \frac{\partial}{\partial t} \left[\frac{\hat{R}}{R^2} \right]_{\text{ret}} &= \frac{1}{c} \frac{\partial}{\partial t} \left[\frac{\hat{R}}{R} \right]_{\text{ret}} - \frac{1}{c} \left[\frac{\hat{R}}{R^2} \right]_{\text{ret}} \frac{\partial}{\partial t} [R]_{\text{ret}} \\
 &= \frac{1}{c} \frac{\partial}{\partial t} \left[\frac{\hat{R}}{R} \right]_{\text{ret}} + \frac{\vec{v} \cdot \hat{R}}{Kc} \left[\frac{\hat{R}}{R^2} \right]_{\text{ret}},
 \end{aligned}$$

$$\begin{aligned}
 \text{then } \vec{E} &= \frac{q}{4\pi\epsilon_0} \left\{ \left[\frac{\hat{R}}{R^2} \right]_{\text{ret}} + \frac{[R]_{\text{ret}}}{c} \frac{\partial}{\partial t} \left[\frac{\hat{R}}{R^2} \right]_{\text{ret}} + \frac{\partial^2}{c^2 \partial t^2} [\hat{R}]_{\text{ret}} \right\} \\
 &= \frac{q}{4\pi\epsilon_0} \left\{ \left[\frac{\hat{R}}{R^2} \right]_{\text{ret}} + \frac{1}{c} \frac{\partial}{\partial t} \left[\frac{\hat{R}}{R} \right]_{\text{ret}} + \frac{\vec{v} \cdot \hat{R}}{Kc} \left[\frac{\hat{R}}{R^2} \right]_{\text{ret}} \right. \\
 &\quad \left. + \frac{1}{c} \frac{\partial}{\partial t} \left[\frac{\hat{R}}{KR} \right]_{\text{ret}} - \frac{1}{c} \frac{\partial}{\partial t} \left[\frac{\hat{R}}{R} \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[\frac{\vec{v}}{KR} \right]_{\text{ret}} \right\} \\
 &= \frac{q}{4\pi\epsilon_0} \left\{ \left[\frac{\hat{R}}{R^2} \right]_{\text{ret}} \left(1 + \frac{\vec{v} \cdot \hat{R}}{Kc} \right) + \frac{1}{c} \frac{\partial}{\partial t} \left[\frac{\hat{R}}{KR} \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[\frac{\vec{v}}{KR} \right]_{\text{ret}} \right\} \\
 &= \frac{q}{4\pi\epsilon_0} \left\{ \left[\frac{\hat{R}}{KR^2} \right]_{\text{ret}} + \frac{1}{c} \frac{\partial}{\partial t} \left[\frac{\hat{R}}{KR} \right]_{\text{ret}} - \frac{1}{c^2} \frac{\partial}{\partial t} \left[\frac{\vec{v}}{KR} \right]_{\text{ret}} \right\}.
 \end{aligned}$$

which is equivalent to Jefimenko's expression.